

# Slack 專用生成式 AI 代理，可透過 API 呼叫執行動作，並回答文件相關問題

來源：<https://codelabs.developers.google.com/genai-for-dev-slack-agent?hl=zh-tw>

步驟數：11

在 Slack 中使用 GenAI Agent，針對文件和 API 呼叫的動作和問答

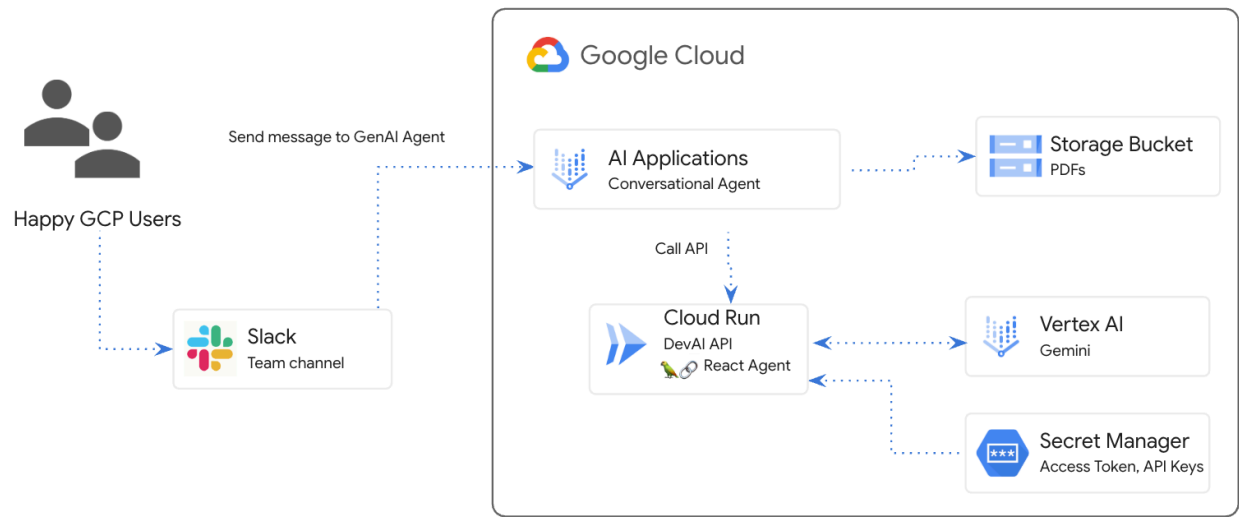
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11. 恭喜！

# 1. 總覽

在本實驗室中，您將建立 GenAI 代理程式、將其連線至 Cloud Run 應用程式，並將代理程式整合至 Slack 工作區。



## 學習目標

本實驗室主要分為幾個部分：

- 部署 Cloud Run 應用程式，與 Gemini API 整合
- 在 AI 應用程式中建立及部署對話式代理
- 將 Agent 整合至 Slack
- 設定資料儲存庫，以便對 PDF 文件進行問答

## 必要條件

- 本實驗室假設您已熟悉 Cloud 控制台和 Cloud Shell 環境。

步驟 2 · 約 0 分鐘

## 2. 設定和需求

### 設定 Cloud 專案

1. 登入 [Google Cloud 控制台](#)，然後建立新專案或重複使用現有專案。如果沒有 Gmail 或 Google Workspace 帳戶，請先[建立帳戶](#)。

The screenshot shows the Google Cloud console interface. At the top, there is a 'Select a project' dropdown menu. Below it, the 'Select a project' page is visible, featuring a search bar labeled 'Search projects and folders'. To the right of the search bar is a 'NEW PROJECT' button. Below the search bar, the 'New Project' form is shown. It includes a 'Project name' field with the text 'My Project 13475' and a help icon. Below the name field is the 'Project ID' field, which contains the text 'inbound-analogy-389614'. A note next to the ID states 'It cannot be changed later.' and there is an 'EDIT' link. Below the ID field is a 'Location' field with a 'BROWSE' button. At the bottom of the form, there is a 'Parent organization or folder' field and two buttons: 'CREATE' and 'CANCEL'.

- **專案名稱**是這個專案參與者的顯示名稱。這是 Google API 未使用的字元字串。你隨時可以更新。
- **專案 ID** 在所有 Google Cloud 專案中都是不重複的，而且設定後即無法變更。Cloud 控制台會自動產生專屬字串，通常您不需要在意該字串為何。在大多數程式碼研究室中，您需要參照專案 ID (通常標示為 `PROJECT_ID`)。如果您不喜歡產生的 ID，可以產生另一個隨機 ID。你也可以嘗試使用自己的名稱，看看是否可用。完成這個步驟後就無法變更，且專案期間會維持不變。
- 請注意，有些 API 會使用第三個值，也就是「專案編號」。如要進一步瞭解這三種值，請參閱[說明文件](#)。

**注意：**專案 ID 在全域中是唯一的，選定後就無法再供他人使用。您是該 ID 的唯一使用者。即使專案已刪除，ID 也無法再次使用

**注意：**如果使用 Gmail 帳戶，可以將預設位置設為「沒有機構」。如果你使用 Google Workspace 帳戶，請選擇適合貴機構的位置。

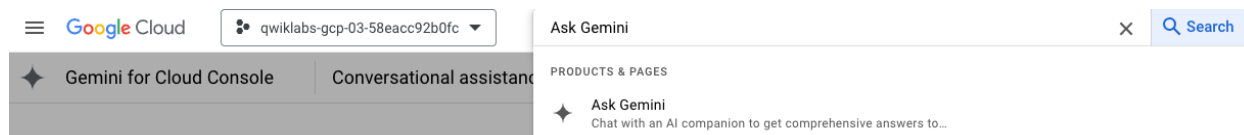
2. 接著，您需要在 Cloud 控制台中[啟用帳單](#)，才能使用 Cloud 資源/API。完成這個程式碼研究室的費用不高，甚至可能完全免費。如要關閉資源，避免在本教學課程結束後繼續產生費用，請刪除您建立的資源或專案。Google Cloud 新使



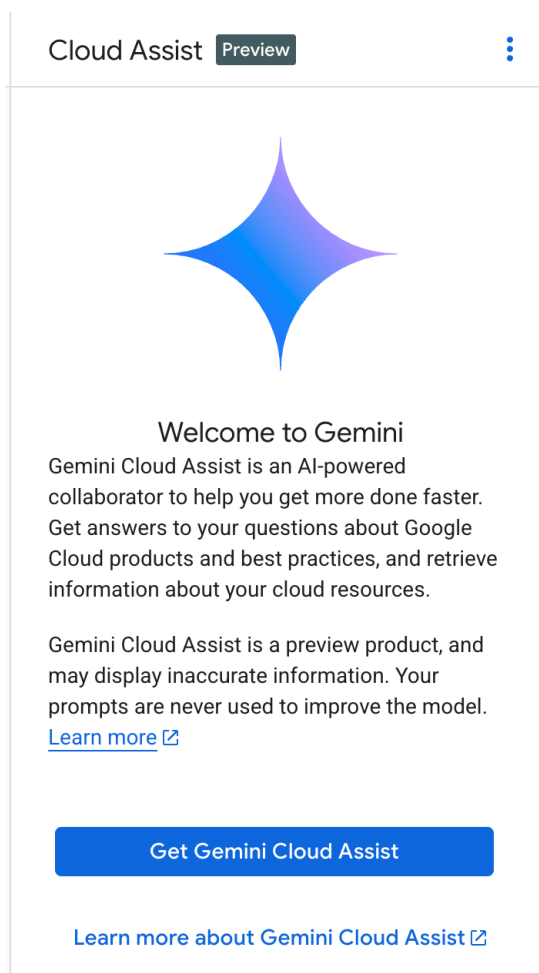
用戶可參加價值\$300 美元的免費試用計畫。

## 環境設定

開啟 Gemini 對話。



啟用 Gemini for Google Cloud API :



Get Gemini Cloud Assist

PREVIEW

✕

Enable Gemini Cloud Assist at no cost by enabling required APIs. Optionally, enable recommended APIs and data sharing for Gemini Cloud Assist. [Learn more about Gemini Cloud Assist](#)

### Required & recommended APIs

1 of 3 recommended APIs is already enabled

2 required APIs and 2 of 2 additional recommended APIs will be enabled

▼

### Terms of Service

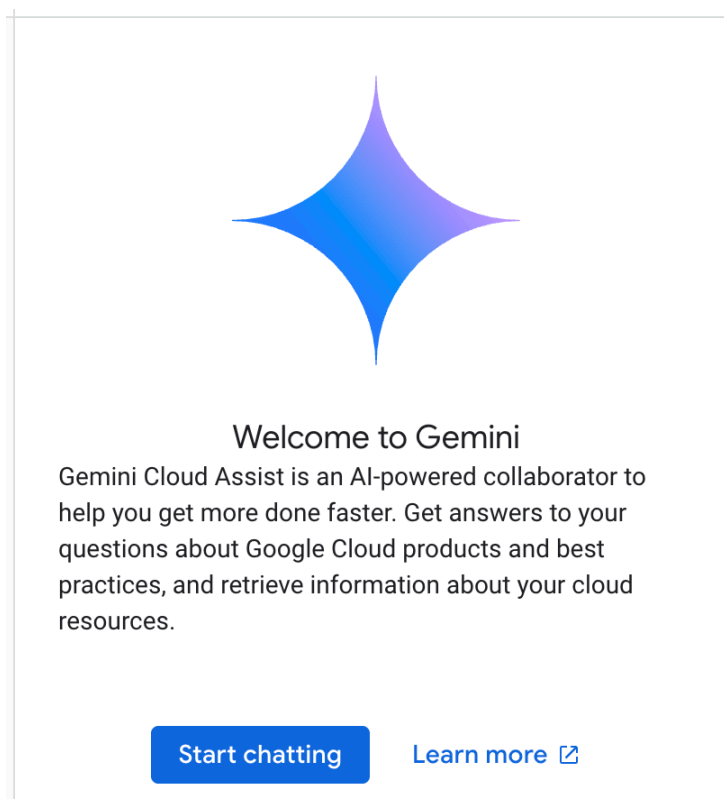
By enabling Gemini Cloud Assist, I acknowledge on behalf of Customer that I have read and agree that Gemini Cloud Assist is subject to the Google Cloud Agreement, including the Pre-GA Offering Terms of the [Google Cloud Platform Specific Service Terms](#), as supplemented by the terms for the [Gemini for Google Cloud Trusted Tester Program](#).

☒ Help Google improve Cloud Assist by sharing your prompts and answers  
Data Use settings can be adjusted at any time after enabling Gemini Cloud Assist

ENABLE GEMINI CLOUD ASSIST AT NO COST

CANCEL

按一下「Start chatting」，然後按照其中一個範例問題操作，或自行輸入提示詞來試用。



建議提示詞：

- 請用 5 個重點說明 Cloud Run。
- 您是 Google Cloud Run 產品經理，請用 5 個簡短重點向學生說明 Cloud Run。

- 您是 Google Cloud Run 產品經理，請用 5 個簡短重點向認證 Kubernetes 開發人員說明 Cloud Run。
- 您是 Google Cloud Run 產品經理，請以 5 個簡短重點，向資深開發人員說明何時該使用 Cloud Run，何時該使用 GKE。

如要進一步瞭解如何撰寫更優質的提示，請參閱[提示指南](#)。

## 瞭解 Gemini for Google Cloud 如何使用您的資料

### Google 對隱私權的承諾

Google 是業界首批發布 [AI/機器學習隱私權承諾](#) 的公司，該文提到我們的信念：除了極致的安全性之外，客戶也應該對儲存在雲端的自家資料保有最大的掌控權。

### 您提交及收到的資料

您向 Gemini 提出的問題 (包括提交給 Gemini 分析或完成的任何輸入資訊或程式碼)，都稱為提示。您從 Gemini 取得的答案或程式碼完成建議稱為「回覆」。我們不會將您的提示詞或 Gemini 的回覆用做模型訓練資料。

### 提示加密

當您將提示提交給 Gemini 時，您的資料會在傳輸過程中加密，然後輸入 Gemini 的基礎模型。

### Gemini 生成的節目資料

Gemini 是以 Google Cloud 第一方程式碼和精選第三方程式碼訓練而成。您必須對程式碼的安全性、測試和效用負責，包括 Gemini 提供的任何程式碼完成、生成或分析功能。

[進一步瞭解](#) Google 如何處理提示。

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步驟 3 · 約 0 分鐘

### 3. 測試提示的選項

你可以透過多種方式測試提示。

- [Vertex AI Studio](#)

Vertex AI Studio 是 Google Cloud Vertex AI 平台的一部分，專門用於簡化及加速生成式 AI 模型的開發和使用。

- [Google AI Studio](#)

Google AI Studio 是網頁工具，可讓您設計提示工程原型，並透過 Gemini API 進行實驗。

- [Gemini 網頁應用程式](#) (gemini.google.com)

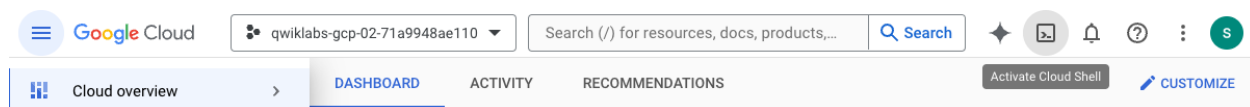
Google Gemini 網頁應用程式 (gemini.google.com) 是一款網頁工具，可協助您探索及運用 Google Gemini AI 模型強大的功能。

- [Android](#)和 [iOS 版 Google 應用程式](#)
-

步驟 4 · 約 0 分鐘

## 4. 建立存放區的本機複本

返回 Google Cloud 控制台，然後點選搜尋列右側的圖示，啟用 Cloud Shell。



如果系統提示您授權，請點選「授權」繼續操作。

### Authorize Cloud Shell

gcloud is requesting your credentials to make a GCP API call.

Click to authorize this and future calls that require your credentials.

REJECT

AUTHORIZE

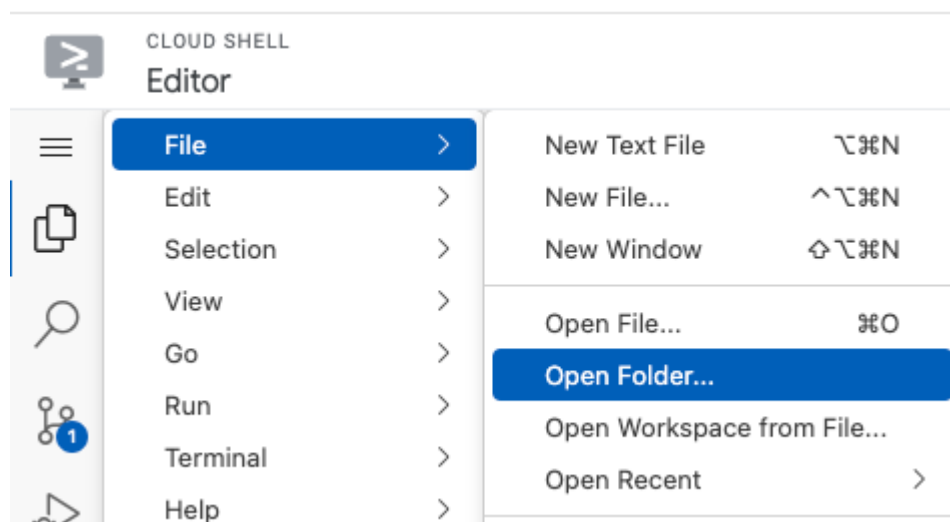
在開啟的終端機中執行下列指令

```
git clone https://github.com/GoogleCloudPlatform/genai-for-developers.git  
  
cd genai-for-developers  
  
git checkout slack-agent-jira-lab
```

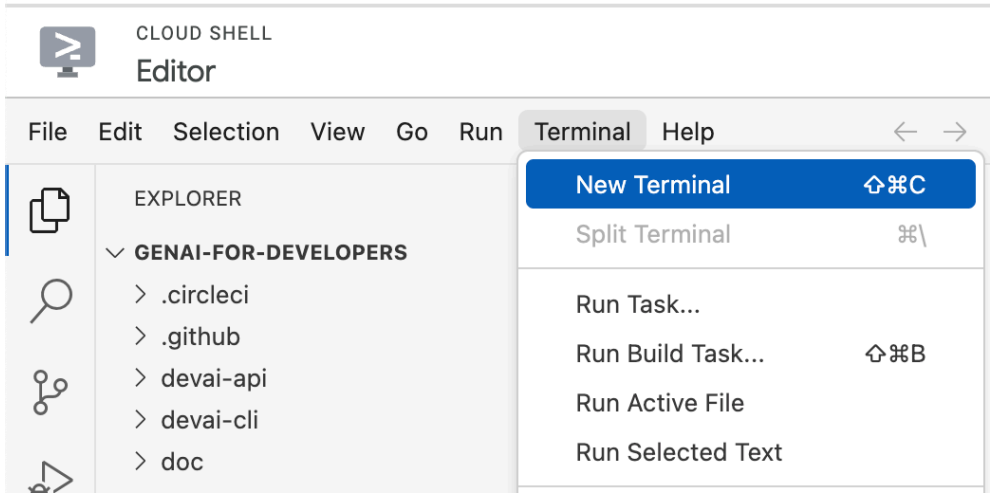
按一下「開啟編輯器」

 Open Editor

使用「File / Open Folder」選單項目開啟「genai-for-developers」。



開啟新的終端機



步驟 5 · 約 0 分鐘

## 5. 建立服務帳戶

建立新的服務帳戶。您將使用這個服務帳戶，從 Cloud Run 應用程式對 Vertex AI Gemini API 進行 API 呼叫。

使用 Qwiklabs 專案詳細資料設定專案詳細資料。

範例：`qwiklabs-gcp-00-2c10937585bb`

```
gcloud config set project YOUR_QWIKLABS_PROJECT_ID
```

建立服務帳戶。

```
export LOCATION=us-central1
export PROJECT_ID=$(gcloud config get-value project)
export SERVICE_ACCOUNT_NAME='vertex-client'
export DISPLAY_NAME='Vertex Client'

gcloud iam service-accounts create $SERVICE_ACCOUNT_NAME --project $PROJECT_ID --display-name
"$DISPLAY_NAME"
```

授予角色。

```
gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/aiplatform.admin"

gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/aiplatform.user"

gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/cloudbuild.builds.editor"

gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/artifactregistry.admin"

gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/storage.admin"

gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/run.admin"

gcloud projects add-iam-policy-binding $PROJECT_ID --
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --
role="roles/secretmanager.secretAccessor"
```

啟用必要服務，即可使用 Vertex AI API 和 Gemini 對話。

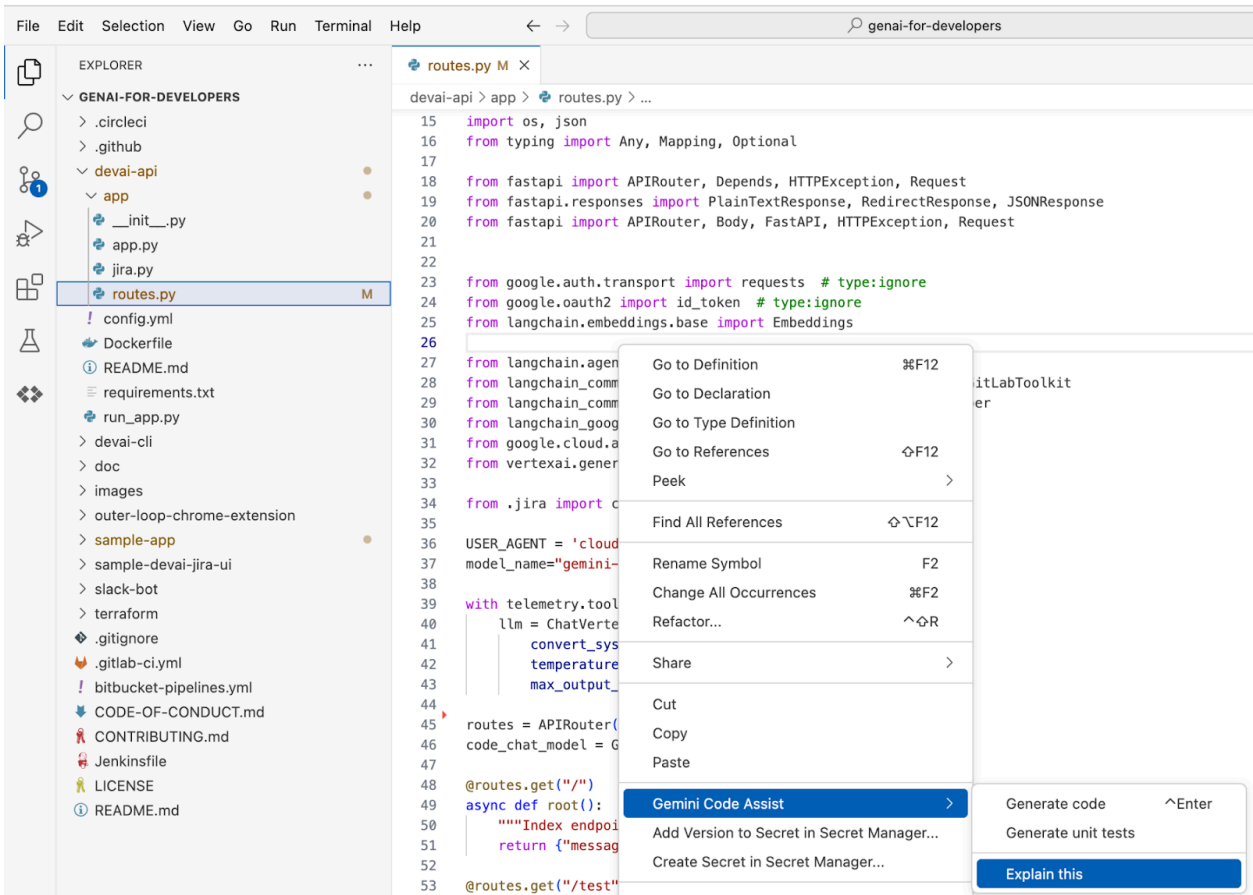
```
gcloud services enable \
  generativelanguage.googleapis.com \
  aiplatform.googleapis.com \
  cloudaicompanion.googleapis.com \
  dialogflow.googleapis.com \
  run.googleapis.com \
  cloudresourcemanager.googleapis.com \
  discoveryengine.googleapis.com
```

啟用必要服務，即可使用 Vertex AI API 和 Gemini 對話。

```
gcloud services enable \
  artifactregistry.googleapis.com \
  cloudbuild.googleapis.com \
  runapps.googleapis.com \
  workstations.googleapis.com \
  servicemanagement.googleapis.com \
  secretmanager.googleapis.com \
  containerscanning.googleapis.com
```

## 使用 Gemini Code Assist 解釋程式碼

開啟「devai-api/app/routes.py」檔案，然後在檔案中的任意位置按一下滑鼠右鍵，並從內容選單中選取「Gemini Code Assist > Explain」this"。





查看 Gemini 對所選檔案的說明。

◆ Gemini

Okay, let's break down the `/home/student_01_fa170799d33c/genai-for-developers/devai-api/app/routes.py` file. This file defines the API routes and their corresponding handlers for a FastAPI application. It leverages Google's Vertex AI (specifically the Gemini model) and interacts with Jira to manage tasks and generate content.

Here's a detailed explanation:

**Overall Purpose**

The primary function of this `routes.py` file is to define the API endpoints that the DevAI application exposes. These endpoints allow users to:

- 1. **Test the API:** A basic `/` and `/test` route
- 2. **Generate content:** The `/generate` route interacts with a large language model (Gemini) to analyze Jira project details and provide a summary of impacted customers and their issues.
- 3. **Create JIRA tickets:** The `/create-jira-issue` route utilizes a large language model (Gemini) to analyze requirements and generates a detailed prompt for a JIRA user story. It then creates a JIRA issue with this prompt.

**Key Components**

**1. Imports:**

- Standard Python libraries: `os`, `json`, `typing`
- FastAPI: `APIRouter`, `Depends`, `HTTPException`, `Request`, `PlainTextResponse`, `RedirectResponse`, `JSONResponse`, `Body`
- Google Cloud: `google.auth.transport.requests`, `google.oauth2.id_token`, `google.cloud.aiplatform.telemetry`
- LangChain: `langchain.embeddings.base.Embeddings`, `langchain.agents`, `langchain_community.agent_toolkits.gitlab.toolkit`, `langchain_community.utilities.gitlab.GitLabAPIWrapper`, `langchain_google_vertexai.ChatVertexAI`
- Vertex AI: `vertexai.generative_models.GenerativeModel`
- Local modules: `.jira` (specifically `create_jira_issue`)

**2. Global Variables:**

- `USER_AGENT`: A string identifier for the application making API calls.

步驟 6 · 約 0 分鐘

## 6. 將 Devai-API 部署到 Cloud Run

確認您位於正確的資料夾。

```
cd ~/genai-for-developers/devai-api
```

在本實驗室中，我們遵循最佳做法，使用 [Secret Manager](#) 在 Cloud Run 中儲存及參照存取權杖和 LangChain API 金鑰值。

設定環境變數。請直接執行這個指令，不要進行任何變更。

```
export JIRA_API_TOKEN=your-jira-token
export JIRA_USERNAME="YOUR-EMAIL"
export JIRA_INSTANCE_URL="https://YOUR-JIRA-PROJECT.atlassian.net"
export JIRA_PROJECT_KEY="YOUR-JIRA-PROJECT-KEY"
export JIRA_CLOUD=true

export GITLAB_PERSONAL_ACCESS_TOKEN=your-gitlab-token
export GITLAB_URL="https://gitlab.com"
export GITLAB_BRANCH="devai"
export GITLAB_BASE_BRANCH="main"
export GITLAB_REPOSITORY="GITLAB-USERID/GITLAB-REPO"

export LANGCHAIN_API_KEY=your-langchain-key
export LANGCHAIN_TRACING_V2=true
export LANGCHAIN_ENDPOINT="https://api.smith.langchain.com"
```

在 Secret Manager 中建立及儲存多個密鑰。

```
echo -n $JIRA_API_TOKEN | \
gcloud secrets create JIRA_API_TOKEN \
--data-file=-

echo -n $GITLAB_PERSONAL_ACCESS_TOKEN | \
gcloud secrets create GITLAB_PERSONAL_ACCESS_TOKEN \
--data-file=-

echo -n $LANGCHAIN_API_KEY | \
gcloud secrets create LANGCHAIN_API_KEY \
--data-file=-
```

將應用程式部署至 Cloud Run。

```
gcloud run deploy devai-api \
  --source=. \
  --region="$LOCATION" \
  --allow-unauthenticated \
  --service-account vertex-client \
  --set-env-vars PROJECT_ID="$PROJECT_ID" \
  --set-env-vars LOCATION="$LOCATION" \
  --set-env-vars GITLAB_URL="$GITLAB_URL" \
  --set-env-vars GITLAB_REPOSITORY="$GITLAB_REPOSITORY" \
  --set-env-vars GITLAB_BRANCH="$GITLAB_BRANCH" \
  --set-env-vars GITLAB_BASE_BRANCH="$GITLAB_BASE_BRANCH" \
  --set-env-vars JIRA_USERNAME="$JIRA_USERNAME" \
  --set-env-vars JIRA_INSTANCE_URL="$JIRA_INSTANCE_URL" \
  --set-env-vars JIRA_PROJECT_KEY="$JIRA_PROJECT_KEY" \
  --set-env-vars JIRA_CLOUD="$JIRA_CLOUD" \
  --set-env-vars LANGCHAIN_TRACING_V2="$LANGCHAIN_TRACING_V2" \
  --update-secrets="LANGCHAIN_API_KEY=LANGCHAIN_API_KEY:latest" \
  --update-secrets="GITLAB_PERSONAL_ACCESS_TOKEN=GITLAB_PERSONAL_ACCESS_TOKEN:latest" \
  --update-secrets="JIRA_API_TOKEN=JIRA_API_TOKEN:latest" \
  --min-instances=1 \
  --max-instances=3
```

回答 ☒ 建立 Artifact Registry Docker 存放區。

Deploying from source requires an Artifact Registry Docker repository to store built containers. A repository named [cloud-run-source-deploy] in region [us-central1] will be created.

Do you want to continue (Y/n)? ☒ y

請 Gemini 說明指令：

What does this command do?

```
gcloud run deploy devai-api \
  --source=. \
  --region="$LOCATION" \
  --allow-unauthenticated \
  --service-account vertex-client \
  --set-env-vars PROJECT_ID="$PROJECT_ID" \
  --set-env-vars LOCATION="$LOCATION" \
  --set-env-vars GITLAB_URL="$GITLAB_URL" \
  --set-env-vars GITLAB_REPOSITORY="$GITLAB_REPOSITORY" \
  --set-env-vars GITLAB_BRANCH="$GITLAB_BRANCH" \
  --set-env-vars GITLAB_BASE_BRANCH="$GITLAB_BASE_BRANCH" \
  --set-env-vars JIRA_USERNAME="$JIRA_USERNAME" \
  --set-env-vars JIRA_INSTANCE_URL="$JIRA_INSTANCE_URL" \
  --set-env-vars JIRA_PROJECT_KEY="$JIRA_PROJECT_KEY" \
  --set-env-vars JIRA_CLOUD="$JIRA_CLOUD" \
  --set-env-vars LANGCHAIN_TRACING_V2="$LANGCHAIN_TRACING_V2" \
  --update-secrets="LANGCHAIN_API_KEY=LANGCHAIN_API_KEY:latest" \
  --update-secrets="GITLAB_PERSONAL_ACCESS_TOKEN=GITLAB_PERSONAL_ACCESS_TOKEN:latest" \
  --update-secrets="JIRA_API_TOKEN=JIRA_API_TOKEN:latest" \
  --min-instances=1 \
  --max-instances=3
```

#### ◆ Gemini

Okay, let's break down the `gcloud run deploy` command you provided. This command is used to deploy a containerized application to Google Cloud Run.

##### In essence, this command does the following:

It deploys the "devai-api" application as a serverless container on Google Cloud Run, using the current directory as the source code, and configures it with several environment variables and secrets for proper operation.

##### Breakdown of the Command:

###### 1. `gcloud run deploy devai-api`

- `gcloud`: This is the Google Cloud command-line interface (CLI).
- `run`: This specifies that we're interacting with Cloud Run.
- `deploy`: This indicates we want to deploy a new service or update an existing one.
- `devai-api`: This is the name we're giving to our Cloud Run service. You can access it using this name (plus the auto generated url).

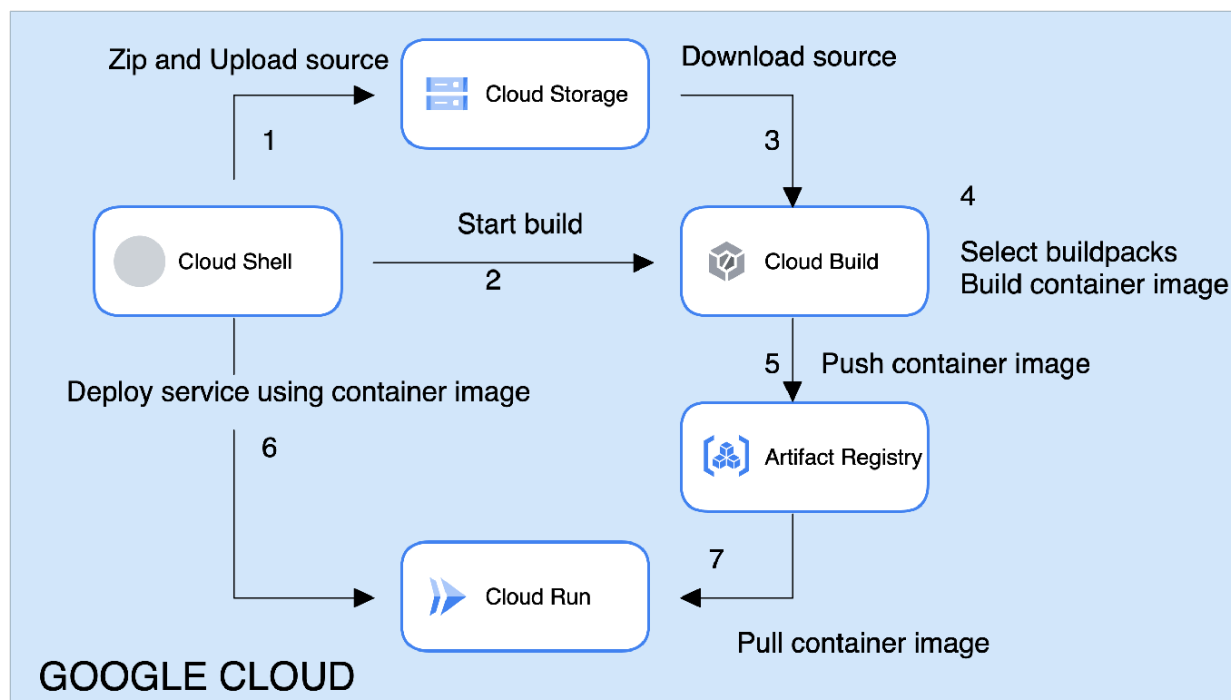
###### 2. `--source=.`

- `--source=.`: This tells Cloud Run to use the current directory (.) as the source code for the application. Cloud Run will build a container image from the files in this directory (presumably including a Dockerfile or Buildpack configuration).

###### 3. `--region="$LOCATION"`

- `--region="$LOCATION"`: This specifies the Google Cloud region where the service should be deployed. The `"$LOCATION"` part means that the region is defined by an environment variable named `LOCATION`. You have to set this env var before running this command.

請參閱下方的 `gcloud run deploy SERVICE_NAME --source=.` 流程。 [瞭解詳情](#)。



在幕後，這項指令會使用 Google Cloud 的 `buildpacks` 和 `Cloud Build`，從您的原始碼自動建構容器映像檔，而不必在機器上安裝 Docker，也不必設定 Buildpacks 或 Cloud Build。也就是說，上述單一指令會執行原本需要 `gcloud builds submit` 和 `gcloud run deploy` 指令才能執行的作業。

如果您已提供 Dockerfile(我們在這個存放區中就是這麼做)，Cloud Build 就會使用該檔案建構容器映像檔，而不是依賴建構套件自動偵測及建構容器映像檔。如要進一步瞭解建構包，請參閱[說明文件](#)。

在 [控制台](#) 中查看 Cloud Build 記錄。

在 [Artifact Registry](#) 中查看建立的 Docker 映像檔。

在 [Cloud Console](#) 中查看 Cloud Run 執行個體詳細資料。

執行 curl 指令來測試端點。

```
curl -X POST \
  -H "Content-Type: application/json" \
  -d '{"prompt": "PROJECT-100"}' \
  $(gcloud run services list --filter="(devai-api)" --format="value(URL)"/generate
```

回覆內容會以 Markdown 格式呈現。以預覽模式顯示，方便閱讀。

## JIRA Project Status Summary - PROJECT-100

**Impacted Customers:**

- **Customer-A:** Experiencing significant issues with new user onboarding and mobile application logging (TEST-M). They are also unable to log in to the system.
- **Customer-B:** Facing delays in issue resolution for existing users and encountering problems accessing desktop application logs (TEST-D). They are also unable to log in to the Liferay portal.

**Issue Breakdown:**

**Common Issues:**

- **Slow Issue Resolution:** Both customers report unacceptable delays in issue resolution, impacting both new and existing users. This suggests a systemic problem in the development or support process.
- **Login Issues:** Both customers report login problems, though the specific applications affected differ. This could indicate a broader authentication issue.

**Customer-Specific Issues:**

- **Customer-A:** Struggling with mobile application (TEST-M) log access, hindering their ability to troubleshoot and monitor the application.
- **Customer-B:** Facing issues with desktop application (TEST-D) log access, similarly impacting their troubleshooting capabilities. They also report problems logging into the Liferay portal, potentially indicating a separate authentication issue specific to that platform.

**Recommendations:**

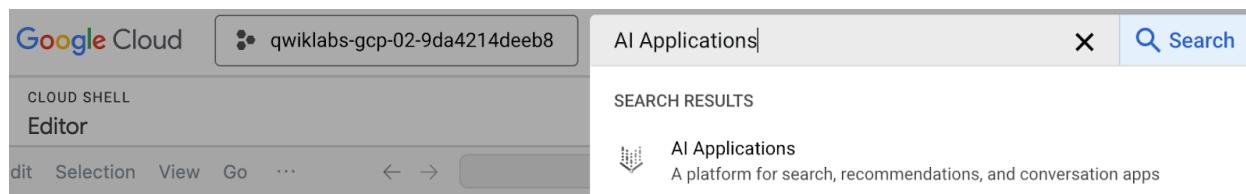
- **Prioritize Issue Resolution:** Implement measures to accelerate issue resolution times, focusing on the backlog affecting both customers.
- **Investigate Login Problems:** Conduct a thorough investigation into the root cause of the login issues, addressing both general authentication concerns and the Liferay portal-specific problem reported by Customer-B.
- **Improve Log Access:** Review and address the issues preventing customers from accessing logs for both the mobile (TEST-M) and desktop (TEST-D) applications.

This summary highlights the critical issues affecting customers in project PROJECT-100. Addressing these issues swiftly and effectively is crucial to ensure customer satisfaction and the project's overall success.

步驟 7 · 約 0 分鐘

## 7. AI Applications

在 Google Cloud 控制台搜尋列中輸入並開啟「AI Applications」。



建立 **Conversational Agent** 應用程式：

### Conversation agents



#### Conversational agent

Build a conversational agent using both rules and generative AI that responds naturally, and predictably.

[CREATE](#) 

選取「Build your own」。

## Get started with Conversational Agents ✕

Please select one of the following options for creating an agent:



### Use a prebuilt agent

Select from a collection of rule-based or generative AI agents tailored for common tasks and scenarios.



### Build your own

Take total control over the experience and leverage powerful AI tools through a simple UI.



### Create a Q&A agent

Provide a URL and we'll create a bot that answers questions based on their content.

輸入顯示名稱的「Agent」，然後按一下「建立」。



Display name\*

Agent

Agent display name

!

You cannot change the [location](#) after creating the agent. Please use the global region for integration with Agent Assist or Insights, as other regions do not support this.

Location\*

us-central1 (Iowa, USA)

Specify a region for your agent

!

This region is not using CMEK [CAI key management](#)

Time zone\*

(GMT-8:00) America/Los\_Angeles

Date and time requests are resolved using this time zone

Default language\*

en — English

The language the agent uses

Conversation start

We can help you get started by defining how your agent will start each conversation.

✦ Playbook

Generative AI

☒ Recommended if you need your agent to start the conversation with natural language tasks such as classifying and steering user requests.

👤 Flow

Rule-based

☐ Recommended if you need your agent to start the conversation with rule-based tasks such as collecting user profile data or a hard-wired welcome message.

Cancel

Create

設定應對手冊名稱：

Agent

設定目標：

Help users with questions about JIRA project

設定指令：



Language  
en

🔍

👤 1

↕

🕒

👤 1

⚙️

<>

💬

|

?

🛡️

🔴 S

Preview: Agent

🐼 📋 ⚙️ ↶️ ⏮️ ↷️

Invocations

+ Save as example

👤 hi

🗣️ Hi there! How can I help you today?

👤 I need to check status of my JIRA project

🗣️ Okay, I can help with that. Could you please tell me the name or key of your JIRA project?

👤 TEST-PROJECT

🗣️ Okay, so you're looking for the status of the TEST-PROJECT. Is that correct?

👤 Yes

🗣️ Great! I'll need a little more information to help you. Could you please tell me what specific information you're looking for about the project? For example, are you looking for the overall project status, the status of a specific task, or something else?

## 代理程式工具設定

將滑鼠游標懸停在左側的扳手圖示上。開啟「工具」選單並建立新工具：

☰

🔧

Conversational Agents

BUILD

📋 Playbooks

👤 Flows

🔧 Tools

📦 Prebuilts

tools by name,

從「類型」下拉式選單中選取 `OpenAPI`。

設定工具名稱：

```
jira-project-status
```

設定說明：

```
Provides JIRA project status
```

切換分頁，檢查 Cloud Run 服務網址的控制台輸出內容。複製服務網址值。

設定結構定義 (YAML) - 取代 YOUR CLOUD RUN URL。

```

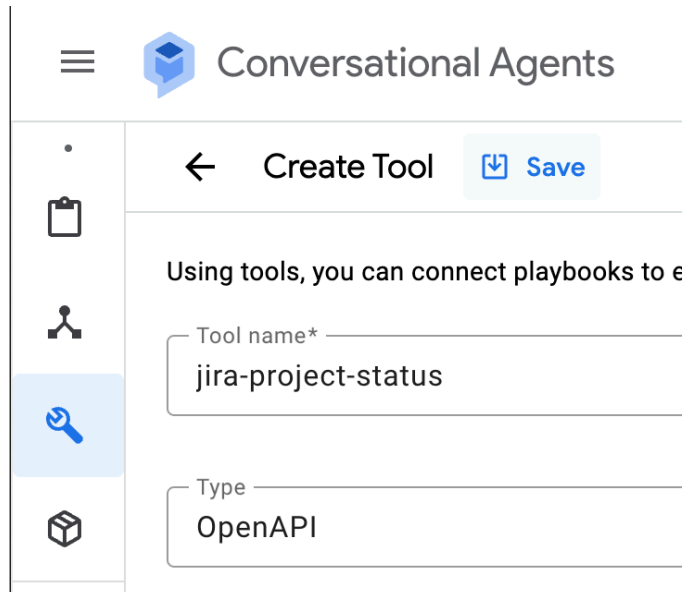
openapi: 3.0.0
info:
  title: CR API
  version: 1.0.0
  description: >-
    This is the OpenAPI specification of a service.
servers:
  - url: 'https://YOUR CLOUD RUN URL'
paths:

  /generate:
    post:
      summary: Request impl
      operationId: generate
      requestBody:
        description: Request impl
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Prompt'
      responses:
        '200':
          description: Generated
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ProjectStatus'

components:
  schemas:
    Prompt:
      type: object
      required:
        - prompt
      properties:
        prompt:
          type: string
    ProjectStatus:
      type: object
      required:
        - message
      properties:
        message:
          type: string

```

儲存工具設定：



Conversational Agents

← Create Tool Save

Using tools, you can connect playbooks to e

Tool name\*

Type

選取左選單中的「Playbooks」，返回 Agent 設定並更新使用工具的指示：

新增使用新工具的操作說明，然後按一下「Save」：

- Use `${TOOL: jira-project-status}` to help the user with JIRA project status.

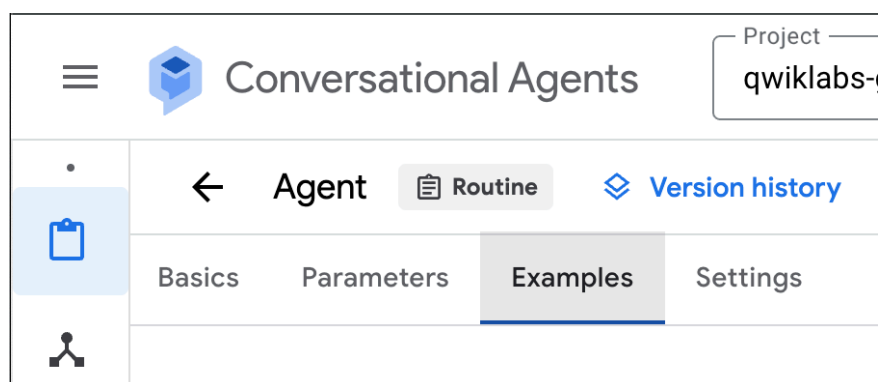
#### Instructions

[Templates](#)

##### Instructions

- 1 - Greet the users, then ask how you can help them today.
- 2 - Summarize the user's request and ask them to confirm that you understood correctly.
- 3 - If necessary, seek clarifying details.
- 4 - Use `${TOOL: jira-project-status}` to help the user with JIRA project status.
- 5 - Thank the user for their business and say goodbye.

切換至「Examples」分頁，然後新增範例：



Conversational Agents

Project

← Agent Routine Version history

Basics Parameters Examples Settings

設定顯示名稱：

jira-project-flow

使用底部的選單，模擬使用者和代理程式之間的對話：

範例流程：

Agent response: What's the project id?

使用者輸入內容：TEST-PROJECT-100

工具使用：jira-project-status

專員回覆：專案狀態詳細資料。

Agent response

Event

Flow details

Flow transition

Playbook invocation

Playbook transition

Retrieved content

Tool use

User input

+

Enter text (@ for other options)

▶

請參考下文建立範例。

✕ Edit example: jira-project-flow

Examples are sample conversations between an end-user and the playbook, including the dialogue and the agent's corresponding actions. [Learn more](#)

Display name\*

jira-project-flow

Basic Settings

⋮



How can I help you today?

⚡

🗑

⋮



I need to review JIRA project status

🗑

⋮



What's the JIRA project id?

⚡

🗑

⋮



TEST-PROJECT-100

🗑

jira-project-status generate

ToolAction

11

Input parametersOutput parameters

Tool\*

jira-project-status

Action

generate

Tool input

☒ Include input

requestBody

Input (requestBody) \*

```
{
  "prompt": "TEST-PROJECT-100"
}
```

Tool output

Output type:

200

Output (200) \*

```
{
  "message": "Project status details for
TEST-PROJECT-100."
}
```

Project status details for TEST-PROJECT-100.

Thank you

Can I help you with anything else?

No, thank you

Agent helped user with project status

Agents support emitting output summary and parameters for exchanging information with other agents.  
[Learn more](#)

End example with output information

Summary\*

Agent helped user with project status

Summary of the Playbook execution result

Conversation state

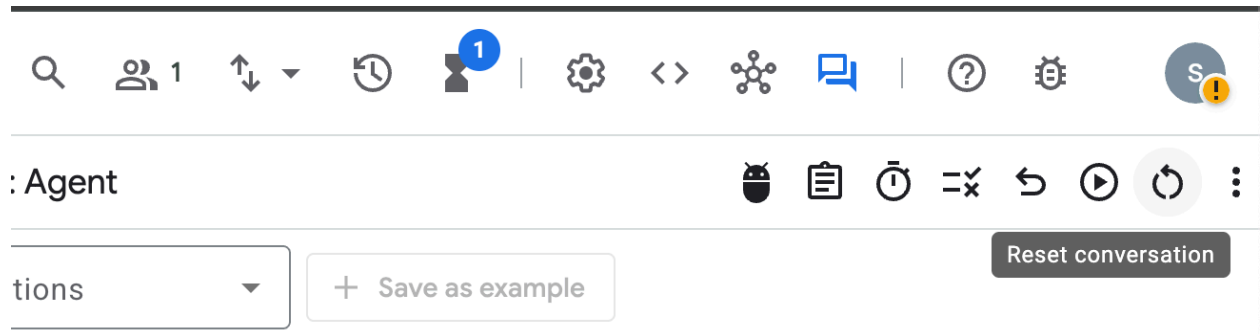
OK

+

Enter text (@ for other options)

按一下 **save**，返回「代理程式模擬器」，然後重設所有現有對話。測試流程。





Invocations ▼

+ Save as example

Agent  
Playbook

hi



Hi there! How can I help you today?



I need to review my project status



Sure, I can help with that. Could you please tell me the project ID?



ABC-100

jira-project-status  
Toolgenerate  
Action1  
Input parameters1  
Output parameters

Okay, here's an analysis of the provided JIRA project status, focusing on impacted customers and their reported issues:

Summary of Impacted Customers and Issues:

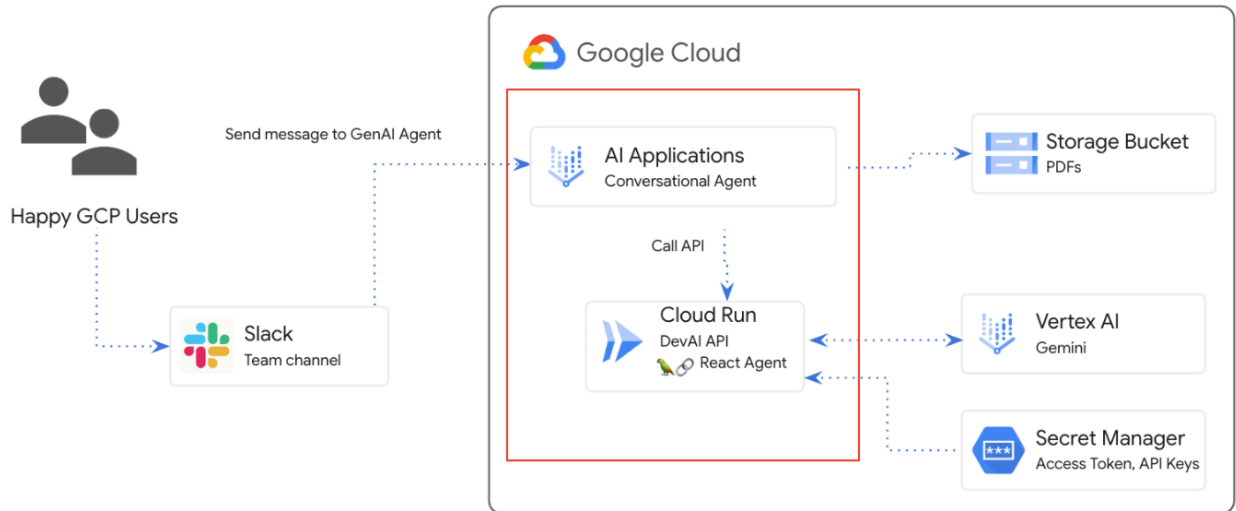
This JIRA project is currently impacting two key customers: Customer-A and Customer-B. Both customers are experiencing significant usability and access issues that need to be addressed.

- Customer-A: This customer is facing problems primarily related to new user experience and basic functionality. The inability to find logs for the mobile application (TEST-M) and the inability to log in are critical blockers. The perception of slow issue resolution further exacerbates the problem.
- Customer-B: This customer is experiencing similar login issues, but also highlights problems impacting existing users and functionality across platforms. The difficulty in finding logs for the desktop application (TEST-D) and login problems with the Liferay portal suggest wider system issues or potentially configuration problems affecting different environments. The slow issue resolution for existing users is also a major concern.

Consolidated Issue Summary (by type):

- Login Issues: Both Customer-A and Customer-B report login problems. This is a critical issue that prevents users from accessing the system and must be prioritized.
- Log Access Issues: Both Customer-A and Customer-B are unable to find their logs for different applications. This makes troubleshooting and debugging difficult, hindering user productivity and potentially masking underlying problems. Specifically affecting mobile application TEST-M for Customer-A and desktop application TEST-D for Customer-B
- Slow Issue Resolution: Both customers are unhappy with the speed at which issues are being resolved. This suggests problems with the support process, development resources, or the complexity of the underlying issues. This slow turn around is impacting both new and existing users

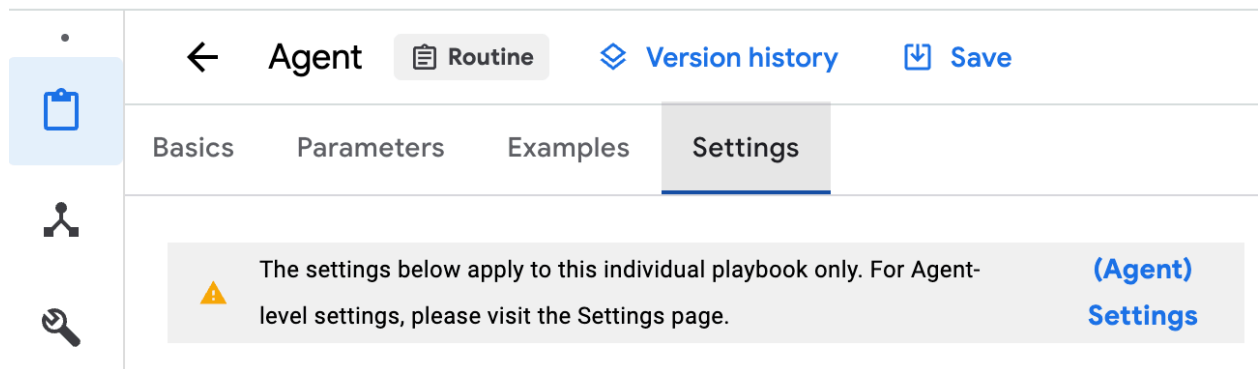
恭喜！整合連結至 Cloud Run 部署應用程式(API) 的工具後，您已成功授權 Agent 執行動作，並擴充 Agent 的功能。



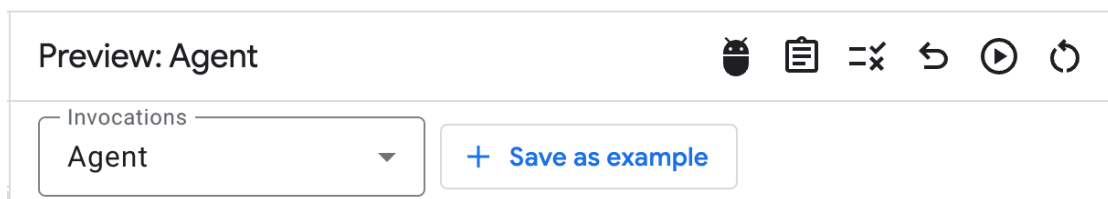
查看 Conversational Agents 的[最佳做法](#)

## 查看可用的代理程式設定

- 記錄設定 - 啟用 Cloud Logging
- Git 整合 - 您可以透過 Git 整合，從 Git 存放區推送及提取代理程式。
- 選擇生成模型
- 詞元限制(輸入和輸出)

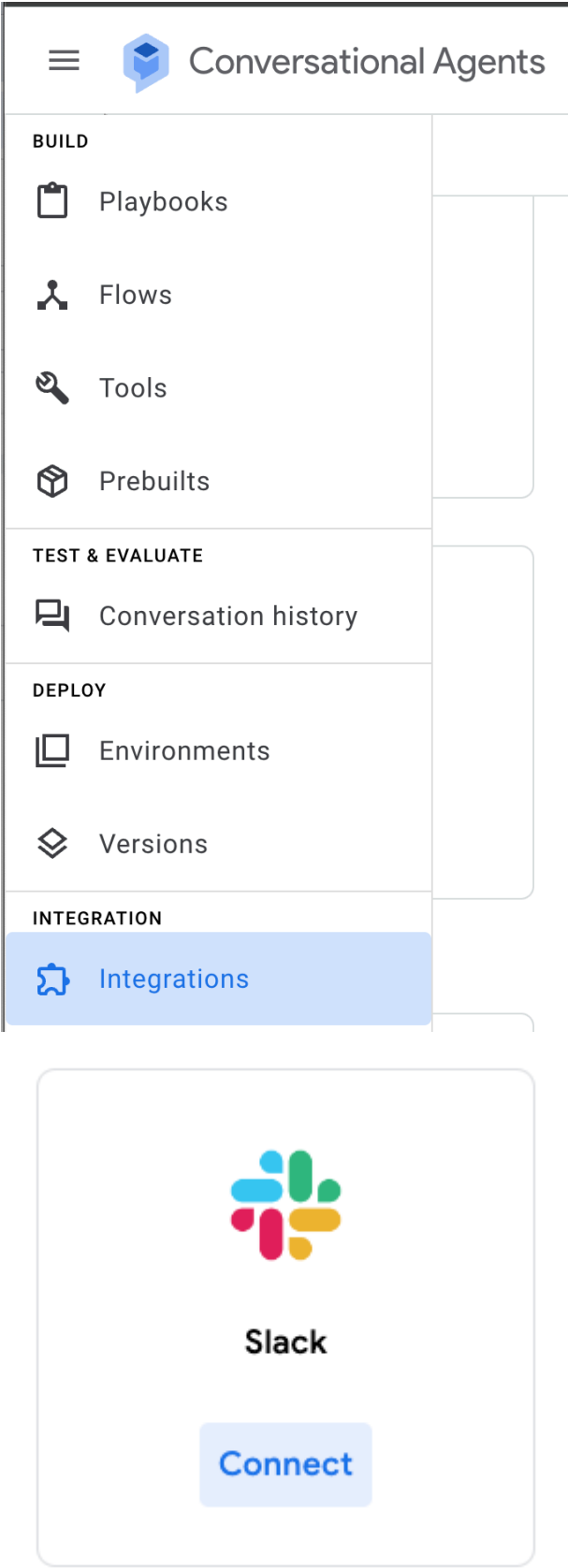


查看代理程式模擬工具控制項：



## 8. Slack 整合

開啟「整合」選單，然後按一下 Slack 圖塊上的「Connect」。



### Slack Integration

Create and teach a conversational bot for Slack.

After you design and test your Conversational agent, you can launch your Slack bot:

1. Get your Slack Access Token and Signing Token and insert them in the fields below.
2. Click 'Start' below.

[View Documentation](#)

#### Configuration

Access token\*

Signing token\*

Environment  
Draft

#### Manage integration

Slack integration is stopped. Start integration to proceed.

Start

Done

開啟連結並在 <https://api.slack.com/apps> 建立新的 Slack 應用程式

 slack api

[Documentation](#) [Tutorials](#) [Your Apps](#)

Your Apps

Create New App

從「資訊清單」中選取：

## Create an app

×

Choose how you'd like to configure your app's scopes and settings.

**From a manifest**

Use a manifest file to add your app's basic info, scopes, settings & features to your app.

>

---

**From scratch**

Use our configuration UI to manually add basic info, scopes, settings, & features to your app.

>

Need help? Check our [documentation](#), or [see an example](#)

選擇要開發應用程式的工作區

## Pick a workspace to develop your app

×

Pick a workspace to develop your app in:

 GenAI-Agents

▼

Keep in mind that you can't change this app's workspace later. If you leave the workspace, you won't be able to manage any apps you've built for it. The workspace will control the app even if you leave the workspace.

[Sign into a different workspace](#)

Step 1 of 3

Cancel

Next

切換至 YAML，然後貼上此資訊清單：

```
display_information:
  name: Agent
  description: Agent
  background_color: "#1148b8"
features:
  app_home:
    home_tab_enabled: false
    messages_tab_enabled: true
    messages_tab_read_only_enabled: false
  bot_user:
    display_name: Agent
    always_online: true
oauth_config:
  scopes:
    bot:
      - app_mentions:read
      - chat:write
      - im:history
      - im:read
      - im:write
      - incoming-webhook
settings:
  event_subscriptions:
    request_url: https://dialogflow-slack-4vnhuutqka-uc.a.run.app
    bot_events:
      - app_mention
      - message.im
  org_deploy_enabled: false
  socket_mode_enabled: false
  token_rotation_enabled: false
```

按一下「建立」：

## Review summary & create your app

Agent

OAuthFeaturesSettings

**Bot Scopes (6)**  
app\_mentions:read, chat:write, im:history, im:read, im:write, incoming-webhook

Step 3 of 3

BackCreate

安裝至 Workspace :

**Basic Information**

## Building Apps for Slack

Create an app that's just for your workspace (or build one that can be used by any workspace) by following the steps below.

---

### Add features and functionality ►

---

### Install your app ▼

Install your app to your Slack workspace to test it and generate the tokens you need to interact with the Slack API. You will be asked to authorize this app after clicking an install option.

**Install to Workspace**

**First, install your app** ✕

This will get your app onto your workspace ►

選取「#general」頻道，然後按一下「允許」





## Agent is requesting permission to access the GenAI-Agents Slack workspace

### What will Agent be able to view?

 Content and info about channels & conversations 

### What will Agent be able to do?

 Perform actions in channels & conversations 

### Where should Agent post?

# Agent requires a channel to post to as an app

# general 

Cancel

Allow

在「Basic Information / App Credentials」下方，複製「Signing Secret」，並在 Agent 的 Slack 整合中，將其設為「Signing Token」欄位的值。

## App Credentials

These credentials allow your app to access the Slack API. They are secret. Please don't share your app credentials with anyone, include them in public code repositories, or store them in insecure ways.

## Signing Secret

●●●●●●●●

Show

Regenerate

Slack signs the requests we send you using this secret. Confirm that each request comes from Slack by verifying its unique signature.

開啟「OAuth & Permissions」，複製「Bot User OAuth Token」，並在 Agent 的 Slack 整合中，將其設為「存取權杖」欄位的值。

## Features

App Home

Org Level Apps

Incoming Webhooks

Interactivity & Shortc...

Slash Commands

Workflow Steps

**OAuth & Permissions**

Event Subscriptions

User ID Translation

App Manifest

Beta Features

## Submit to App Directory

Review & Submit



At least one redirect URL needs to be set below  
can be opted into token rotation

Opt In

## OAuth Tokens for Your Workspace

These tokens were automatically generated when you  
your team. You can use these to authenticate your app

### Bot User OAuth Token

xoxb-7061551383478-7253488939541-e7y1Y8fRl

Access Level: Workspace

Reinstall to Workspace

設定必填欄位，然後按一下「開始」。

代理程式的「Access Token」(存取權杖) 值是 Slack 的「Bot User OAuth Token」(機器人使用者 OAuth 權杖)。

代理程式的「簽署權杖」值是 Slack 的「簽署密鑰」。

## Slack Integration

Create and teach a conversational bot for Slack.

After you design and test your Conversational agent, you can launch your Slack bot:

1. Get your Slack Access Token and Signing Token and insert them in the fields below.
2. Click 'Start' below.

[View Documentation](#)

### Configuration

Access token\*

xoxb-7061551383478-8590862076273-o1OnjBgbk000000000

Signing token\*

5cc7a9b7134d12b268c8ca0000000000

Environment

Draft

### Manage integration

Slack integration is stopped. Start integration to proceed.

Start

Done

## Webhook URL

Webhook URL

<https://dialogflow.cloud.google.com/v1/cx/locations/us-central1/integrations>



## Manage integration

Slack integration is started.

Stop

複製「Webhook 網址」，然後返回 Slack 應用程式設定。

開啟「事件訂閱」部分，然後貼上網址。

Dialogflow ...

Settings

Basic Information

Collaborators

Socket Mode

Install App

Manage Distribution

Features

App Home

Org Level Apps

Incoming Webhooks

Interactivity & Shortc...

Slash Commands

Workflow Steps

OAuth & Permissions

Event Subscriptions

User ID Translation

App Manifest

Beta Features

## Event Subscriptions

### Enable Events

On

Your app can subscribe to be notified of events in Slack (for example, when a user adds a reaction or creates a file) at a URL you choose. [Learn more.](#)

**Request URL** Verified ✓

https://dialogflow.cloud.google.com/v1/cx/locations/us-centra

Change

We'll send HTTP POST requests to this URL when events occur. As soon as you enter a URL, we'll send a request with a `challenge` parameter, and your endpoint must respond with the challenge value. [Learn more.](#)

### New event authorization format

i

**Recent changes to Events API payloads**

The Events API now sends information about authorized users and workspaces in a new, compact format. [Learn more.](#)

儲存變更。

Discard Changes

Save Changes

開啟「Slack」，然後輸入「@Agent」新增代理。

例如新增名為「@CX」的應用程式。

CX APP

Not in channel

Andrey Shakirov (you)

@channel Notify everyone in this channel.

@here Notify every online member in this channel.

@

+

Aa

😊

@

📷

🎤

📝



**Andrey Shakirov** 5:52 AM

@CX

Only visible to you



**Slackbot** 5:52 AM

You mentioned @CX, but they're not in this private channel.

Invite Them

Do Nothing

## Want to add this person instead?



@CX isn't in # genai-agents yet — you can add them now instead of sending this message to the whole channel.

Learn more about adding people to channels by visiting our [Help Center](#).

Send This Message

Add to Channel

請專員提供 JIRA 專案摘要。

🕒

Search GenAI-Agents

?

🔒 jira-epic-testing ▾

👤 1

🔊

🖼️ Canvas

+ Add a bookmark

Today ▾

👤

Andrey Shakirov

5:52 AM

@CX

👁️ Only visible to you

🤖

Slackbot

5:52 AM

OK! I've invited @CX to this channel.

👤

CX APP

5:52 AM

was added to jira-epic-testing by Andrey Shakirov.

👤

CX APP

5:52 AM

Hi, how can I help you today?

👤

Andrey Shakirov

5:52 AM

@CX

Hello, I need to get summary for my JIRA project

👤

CX APP

5:52 AM

What is the project ID?

👤

Andrey Shakirov

5:53 AM

@CX

INTEGRATIONS-275

👤

CX APP

5:53 AM

Here is the summary for INTEGRATIONS-275: "## JIRA Project Status Summary\n\n\*\*Impacted Customers:\*\*\n\n\* \*\*Customer-A:\*\* Experiencing significant issues with new user onboarding, log access for the TEST-M mobile application, and general login functionality. \n\n\* \*\*Customer-B:\*\* Facing delays in issue resolution for existing users, encountering problems accessing logs for the TEST-D desktop application, and experiencing login issues specifically with the Liferay portal.\n\n\*\*Reported Issues & Analysis:\*\*\n\n| Issue | Customer | Description | Impact | Potential Root Cause (Speculative) |\n\n-----|-----|-----|-----|-----|

-----\n\n| Issue-1 | A & B | Delays in issue resolution for new/existing users | Reduced user productivity, potential project delays, increased support burden. | Resource allocation, process bottleneck, technical debt |\n\n| Issue-2 | A | Can't find logs for mobile app TEST-M | Hinders debugging and issue resolution for TEST-M users. | Logging configuration error, storage issue, permission problem |\n\n| Issue-2 | B | Can't find logs for desktop app TEST-D | Hinders debugging and issue resolution for TEST-D users. | Logging configuration error, storage issue, permission problem |\n\n| Issue-3 | A | Can't login | Blocks all functionality for Customer A, potentially impacting all users. | Authentication service down, misconfiguration, network issue |\n\n| Issue-3 | B | Can't login into Liferay portal | Blocks access to Liferay for Customer B, impacting users dependent on this portal. | Liferay-specific configuration issue, authentication problem |\n\n\*\*Requirements:\*\*\n\n\* \*\*INTEGRATIONS-275:\*\* The provided information does not clarify the relationship of this requirement to the reported issues. Further information is needed to analyze its impact.\n\n\*\*Recommendations:\*\*\n\n\* \*\*Prioritize Issue-1 and Issue-3:\*\* These issues significantly impact user productivity and access. Investigating the root causes and implementing immediate mitigations is crucial.\n\n\* \*\*Investigate Logging Issues (Issue-2):\*\* Identify

B

I

🔗

🔗

🔗

🔗

🔗

🔗

🔗

🔗

Message @jira-epic-testing

+

Aa

😊

@

📎

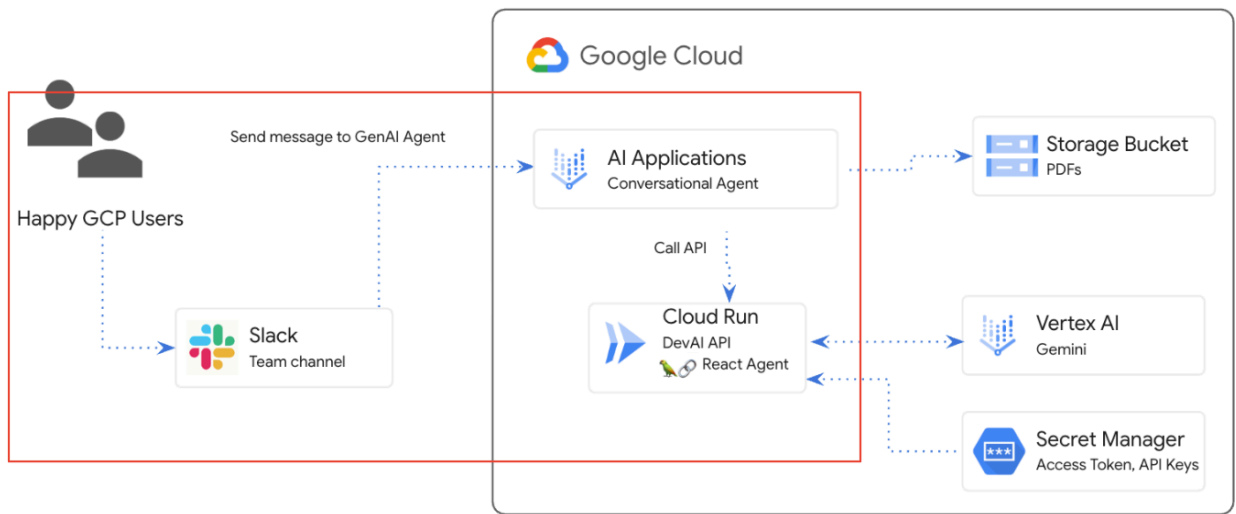
🔊

🔗

➡

▾

恭喜！代理程式已成功整合至 Slack 工作區。



步驟 9 · 約 0 分鐘

## 9. 詢問 PDF 文件內容

本節說明如何使用 PDF 文件建立資料儲存庫，並將其連結至 Agent，根據文件內容啟用問與答功能。

### 建立 Cloud Storage bucket

開啟 Cloud Shell：<https://shell.cloud.google.com/>

使用 GCP 專案的最後 5 位數設定 bucket 名稱。範例：`pdf-docs-3dw21`

```
BUCKET_NAME=pdf-docs-LAST-5-DIGITS-OF-YOUR-PROJECT
```

建立 bucket 並上傳 PDF 文件。

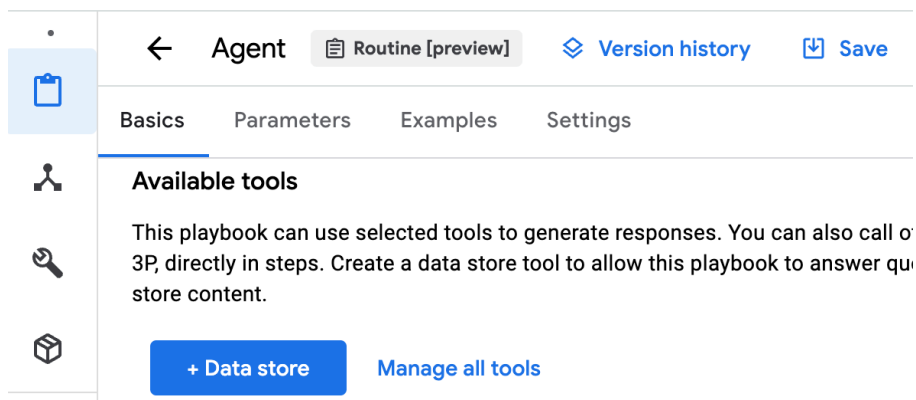
```
gcloud storage buckets create gs://$BUCKET_NAME \
  --location=us \
  --default-storage-class=STANDARD \
  --no-public-access-prevention \
  --uniform-bucket-level-access

wget https://services.google.com/fh/files/misc/exec_guide_gen_ai.pdf

gsutil cp exec_guide_gen_ai.pdf gs://$BUCKET_NAME
```

### 資料儲存庫設定

返回代理控制台並開啟「Agent」，向下捲動並點選「+ Data store」。



請使用下列值：

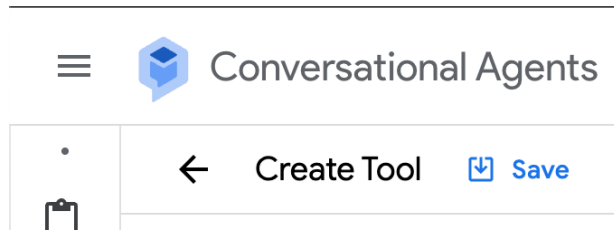
工具名稱：`pdf-docs`

類型：`Data store`

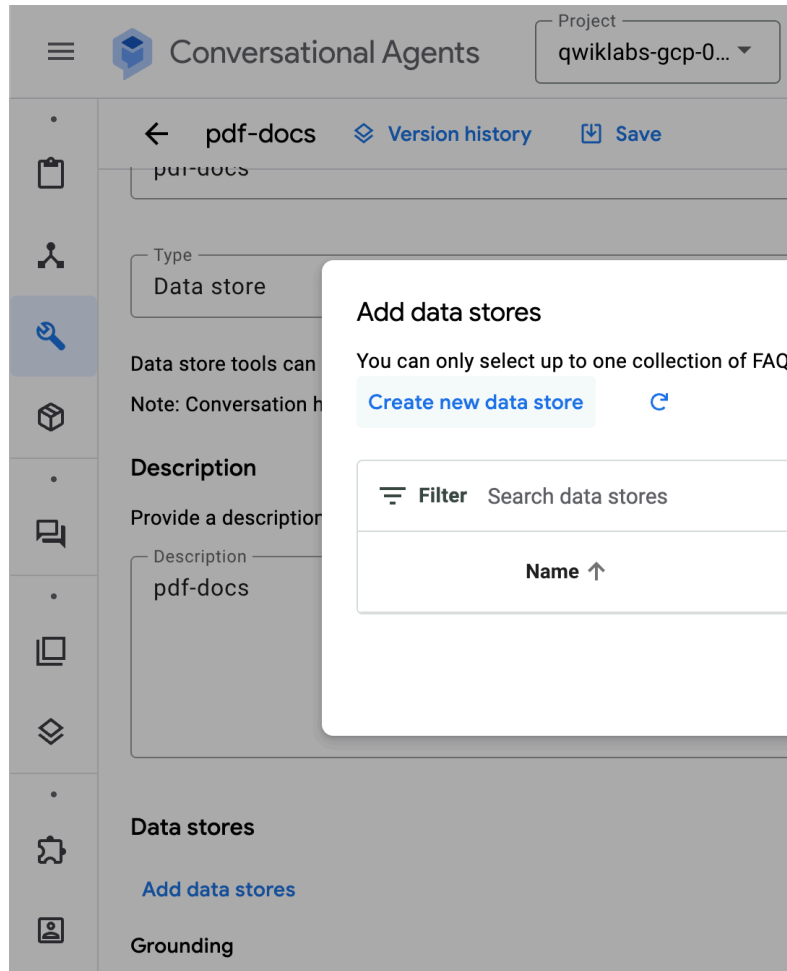
說明：`pdf-docs`

按一下「Save」





按一下頁面底部的「Add data stores」，然後按一下「Create new data store」。



選取「Cloud Storage」做為資料來源。

選取：Unstructured documents

然後選取 GCS bucket/資料夾。

← Create Data Store

✓ Source

2 Data

3 Configuration

Import data from Cloud Storage

Specify the data you want to import to your data store

What kind of data are you importing?

For more information, see [Prepare data for ingesting](#).

☒ Unstructured documents (PDF, HTML, TXT and more)

Supported file formats: PDF, HTML, TXT (CSV for FAQ, DOCX and PPTX are available)

☐ Structured data (JSONL)

General structured data in JSON format. The schema is auto-detected

☐ Enhanced structured data (JSONL with document IDs)

The file must follow the specific root-level schema. You must provide an ID for each document.

☐ Structured media data (JSONL containing media files)

Structured media files. [View requirements](#)

☐ Linked unstructured documents (JSONL with metadata)

JSONL files with unstructured document links and its metadata. The schema is auto-detected

☐ Structured FAQ data for a chat application (CSV)

Structured FAQ data

☐ Unstructured documents for Healthcare

[PREVIEW](#)

(PDF, XML and more)

Synchronization frequency

This action cannot be changed after the data store is created

☒ One time

One time ingestion only

☐ Periodic

Periodic data ingestion

Select a folder or a file you want to import

FOLDER

FILE

gs:// \*

☒ pdf-docs-367bc

CONTINUE

CANCEL

Select Cloud Storage location.

← Buckets ▾

pdf-docs-367bc

選取資料儲存庫位置的「us」。

資料儲存庫名稱請輸入「pdf-docs」

從下拉式選單中選取「Digital parser」。

啟用進階分塊。

在分塊中啟用祖系標題。

請按一下 [ Create ]。

## Configure your data store

Configure additional settings for your data store

### Location of your data store

Multi-region

us (multiple regions in United States)

### Your data store name

Data store name \*

pdf-docs

ID: pdf-docs\_1718877956937. It cannot be changed later. [EDIT](#)

☐ This data store contains access control information

### DOCUMENT PROCESSING OPTIONS

#### Document parsing

Select a parser to apply to your documents. [Learn more about parsing and chunking documents.](#)

Default document parser \*

Digital Parser

Select the parser override for specific document formats:

[+ ADD EXCEPTION FOR A FILE FORMAT](#)

#### Document chunking

You can't change document chunking after creating the data store

☒ Enable advanced chunking configuration

Chunk size limit

500

Tokens

☒ Include ancestor headings in chunks

CREATE

CANCEL

You have successfully created a data store



點選資料儲存庫，然後查看「文件」、「活動」和「處理設定」。

DOCUMENTS

ACTIVITY

PROCESSING CONFIG

PREVIEW

Activity ?

[+ IMPORT DATA](#)

Filter

| Status             | Detail    | Items succeeded | Operation name                       |
|--------------------|-----------|-----------------|--------------------------------------|
| Import in progress | No errors | 0               | import-documents-7053821392880734941 |

匯入作業大約需要 5 到 10 分鐘。

DOCUMENTS

ACTIVITY

PROCESSING CONFIG

PREVIEW

Activity

+ IMPORT DATA

Filter

| Status             | Detail    | Items succeeded | Operation name                       |
|--------------------|-----------|-----------------|--------------------------------------|
| ✔ Import completed | No errors | 1               | import-documents-7053821392880734941 |

剖析和分塊選項

您可以透過下列方式控制內容剖析：

- **數位剖析器**。除非指定其他剖析器類型，否則所有檔案類型預設都會使用數位剖析器。如果資料儲存庫未指定其他預設剖析器，或指定的剖析器不支援擷取文件的檔案類型，數位剖析器就會處理擷取的文件。
- **PDF 的 OCR 剖析**。公開預先發布版。如果您打算上傳掃描的 PDF 或內含圖片文字的 PDF，可以開啟 OCR 剖析器，提升 PDF 索引品質。請參閱「關於 PDF 的 OCR 剖析功能」。
- **版面配置剖析器**。如果您打算將 Vertex AI Search 用於 RAG，請為 HTML、PDF 或 DOCX 檔案啟用版面配置剖析器。如要瞭解這個剖析器和如何啟用，請參閱「為 RAG 分塊文件」。

進一步瞭解如何剖析文件及進行分塊。

工具設定

返回「代理程式工具」設定畫面，然後重新整理可用的資料儲存庫。

選取剛建立的資料儲存庫，然後按一下「Confirm」。

Add data stores

You can only select up to one collection of FAQ, website, and unstructured data store type.

Create new data store

1 data store selected.

Filter

Search data stores

|   | Name ↑   | Create Time  | Type                   |
|---|----------|--------------|------------------------|
| ✔ | pdf-docs | Mar 18, 2025 | Unstructured documents |

Confirm

Cancel

設定依據。

輸入「Google Cloud」做為公司名稱。

酬載設定 - 勾選「Include snippets in Conversational Messenger response payload」。將值設為 5。

按一下「儲存」。

☰

Conversational Agents

Project  
qwiklabs-gcp-0... ▾

•

📄

👤

🔍

📦

•

💬

•

📄

📦

•

⚙️

👤

➤

← pdf-docs 🔗 Version history 📄 Save

Data stores

[Add data stores](#)

Filter Search data stores

Name ↑

pdf-docs (processing, available in 10 mins)

Grounding

Grounding gauges the confidence that all information in the response is

☒ Enable grounding

Each response generated from the content of your connected data stores is given a score

Lowest score allowed

Low: We have low confidence that the response is gr

☒ Apply grounding heuristics filter

We've created this extra filter to catch and suppress responses containing content

Company name

Company

Google Cloud

Data store model selection

Select generative model

Default

[Learn more about model versioning](#)

Summarization prompt

Text that is sent to the LLM, that can contain predefined placeholders. N

☒ Use the default prompt

☐ Create a custom prompt

Payload settings

☒ Include snippets in Conversational Messenger response payload

Maximum number of snippets in payload (1-5)

5

## 設定代理程式的指令

返回代理程式設定。

新增指令：

```
- Provide detailed answer to users questions about the exec guide to gen ai using information in the ${TOOL:pdf-docs}
```

Instructions🔗 Templates

Instructions

1

- Greet the users, then ask how you can help them today.

2

- Summarize the user's request and ask them to confirm that you understood correctly.

3

- If necessary, seek clarifying details.

4

- Use `${TOOL:jira-project-status}` to help the user with JIRA project status.

5

- Provide detailed answer to users questions about the exec guide to gen ai using information in the `${TOOL:pdf-docs}`

6

- Thank the user for their business and say goodbye.

儲存設定。

為 PDF 文件工具建立範例

切換至「範例」分頁。建立新範例：`Guide to generative AI`

使用「+」動作：

Agent response

Event

Flow transition

Playbook invocation

Playbook transition

Retrieved content

Tool use

User input

+

Enter text (@ for other options)

▶

新增「使用者輸入內容」：

```
What are the main capabilities for generative AI?
```

新增「工具使用」功能。

- 工具和動作：「pdf-docs」

輸入 (requestBody)

```
{
  "query": "Main capabilities for generative AI",
  "filter": "",
  "userMetadata": {},
  "fallback": ""
}
```

工具輸出內容：

```
{
  "answer": "Detailed answer about main capabilities for generative AI",
  "snippets": [
    {
      "uri": "https://storage.cloud.google.com/pdf-docs-49ca4/exec_guide_gen_ai.pdf",
      "text": "Detailed answer about main capabilities",
      "title": "exec_guide_gen_ai"
    }
  ]
}
```

新增「Agent response」

Detailed answer about main capabilities.

[https://storage.cloud.google.com/pdf-docs-49ca4/exec\\_guide\\_gen\\_ai.pdf](https://storage.cloud.google.com/pdf-docs-49ca4/exec_guide_gen_ai.pdf)

設定範例：

Display name\*

Guide to generative AI

### Basic Settings

What are the main capabilities for generative AI?

pdf-docs pdf-docs 1 1  
Tool Action Input parameters Output parameters

Detailed answer about main capabilities.  
[https://storage.cloud.google.com/pdf-docs-49ca4/exec\\_guide\\_gen\\_ai.pdf](https://storage.cloud.google.com/pdf-docs-49ca4/exec_guide_gen_ai.pdf)

Agent answered questions based on the PDF document

工具叫用設定：

pdf-docs pdf-docs 1 1  
Tool Action Input parameters Output parameters

Tool\*  
pdf-docs

Action  
pdf-docs

Tool input

☒ Include input requestBody

Input (requestBody) \*

```
{
  "fallback": "",
  "query": "Main capabilities for generative AI",
  "filter": "",
  "userMetadata": {}
}
```

View Output

Tool output

Output type: 200

Output (200) \*

```
{
  "answer": "Detailed answer about main capabilities for generative AI",
  "snippets": [
    {
      "text": "Detailed answer about main capabilities for generative AI",
      "title": "exec_guide_gen_ai",
      "uri": "https://storage.cloud.google.com/pdf-docs-49ca4/exec_guide_gen_ai.pdf"
    }
  ]
}
```

切換至模擬器，測試設定。

問題：



What are the 10 steps in the exec guide?

## Preview: Agent



Invocations

+ Save as example



What are the 10 steps in the exec guide?



pdf-docs  
Tool

pdf-docs  
Action

1  
Input parameters

1  
Output parameters



The 10 steps to launch your first generative AI use case in 30 days are:



1. Identify a specific domain
2. Create a tiger team
3. Select a persona
4. Determine data sources
5. Build UX and UI
6. Design prompts with tiger team
7. Expand to more users
8. Define your goals
9. Build an LM operations plan
10. Expand to more use cases

從「呼叫」下拉式選單中選取「Agent」，然後按一下「Save as example」。

## Preview: Agent

Invocations

Agent

+ Save as example



What are the 10 steps in the exec guide?



pdf-docs  
Tool

pdf-docs  
Action

1  
Input parameters

The 10 steps to launch your first generative AI use case in 30 days are:



1. Identify a specific domain
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4. Determine data sources
5. Build UX and UI
6. Design prompts with tiger team
7. Expand to more users
8. Define your goals
9. Build an LM operations plan
10. Expand to more use cases

提供名稱「user-question-flow」。

設定摘要：「Agent helped user answer question based on the pdf document」，然後按一下「儲存」。

設定服務專員回覆格式，並在工具輸出內容部分加入 PDF 文件連結。

Display name\*

user-question-flow

Basic Settings

What are the 10 steps in the exec guide?

pdf-docs

Tool

pdf-docs

Action

1

Input parameters

1

Output parameters

The 10 steps to launch your first generative AI use case in 30 days are:

1. Identify a specific domain

2. Create a tiger team

3. Select a persona

4. Determine data sources

5. Build UX and UI

6. Design prompts with tiger team

7. Expand to more users

8. Define your goals

9. Build an LM operations plan

10. Expand to more use cases

https://storage.cloud.google.com/pdf-docs-367bc/exec\_guide\_gen\_ai.pdf

Jump to bottom content%20The,in%2030%20days

Agent helped user answer question based on the pdf document

Agents support emitting output summary and parameters for exchanging information with other agents.

[Learn more](#)

End example with output information

Summary\*

Agent helped user answer question based on the pdf document

Summary of the Playbook execution result

Conversation state

OK

儲存範例。

返回模擬工具並點按「Replay conversation」，查看更新後的回覆格式。如果在儲存範例後看到錯誤訊息，可能需要重新整理瀏覽器視窗，然後再次傳送提示來重新啟動對話：

What are the 10 steps in the exec guide?

Preview: Agent



Invocations

+ Save as example



What are the 10 steps in the exec guide?



pdf-docs  
Tool



pdf-docs  
Action

1  
Input parameters

1  
Output parameters



The 10 steps to launch your first generative AI use case in 30 days are:

1. Identify a specific domain
2. Create a tiger team
3. Select a persona
4. Determine data sources
5. Build UX and UI
6. Design prompts with tiger team
7. Expand to more users
8. Define your goals
9. Build an LM operations plan
10. Expand to more use cases

[https://storage.cloud.google.com/pdf-docs-367bc/exec\\_guide\\_gen\\_ai.pdf#:~:text=Generate%20content%20The,in%2030%20days](https://storage.cloud.google.com/pdf-docs-367bc/exec_guide_gen_ai.pdf#:~:text=Generate%20content%20The,in%2030%20days)



Agent (OK)  
Playbook

提出其他問題：

What are the main capabilities in the exec guide?



What are the main capabilities in the exec guide?



pdf-docs  
Tool



pdf-docs  
Action

1  
Input parameters

1  
Output parameters



The executive guide to generative AI covers the following main capabilities:

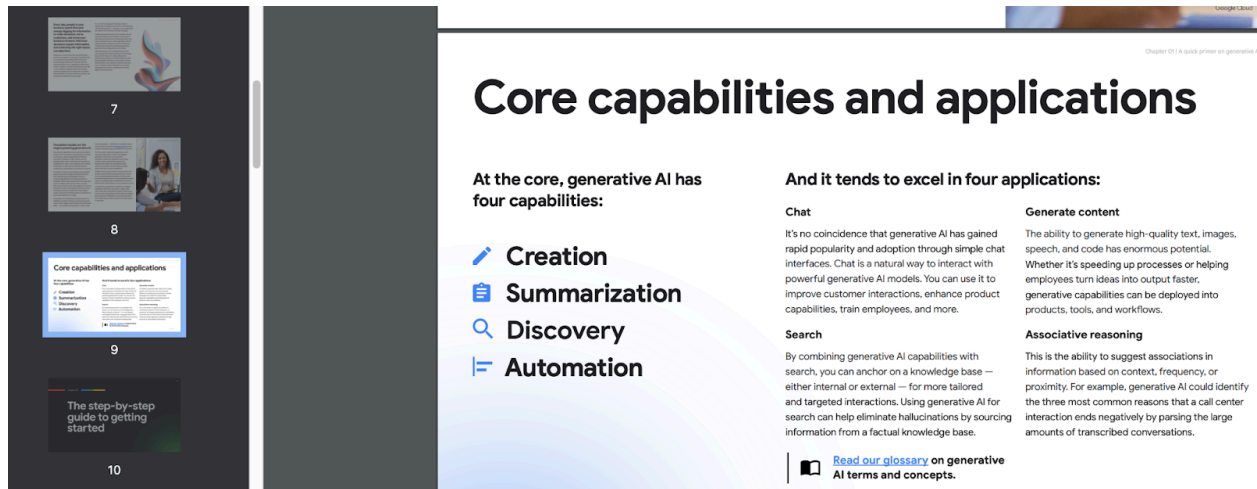
- A quick primer on generative AI
- A step-by-step guide to getting started with generative AI
- The value of generative AI in every industry

[https://storage.cloud.google.com/pdf-docs-367bc/exec\\_guide\\_gen\\_ai.pdf#:~:text=In%20the%20first,powering%20generative%20AI](https://storage.cloud.google.com/pdf-docs-367bc/exec_guide_gen_ai.pdf#:~:text=In%20the%20first,powering%20generative%20AI)



Agent (OK)  
Playbook

來源 PDF 文件。



問題：

What should I consider when evaluating projects?

 What should I consider when evaluating projects?

 pdf-docs  
Tool

pdf-docs  
Action

1  
Input parameters

1  
Output parameters

When evaluating generative AI projects, consider the following:

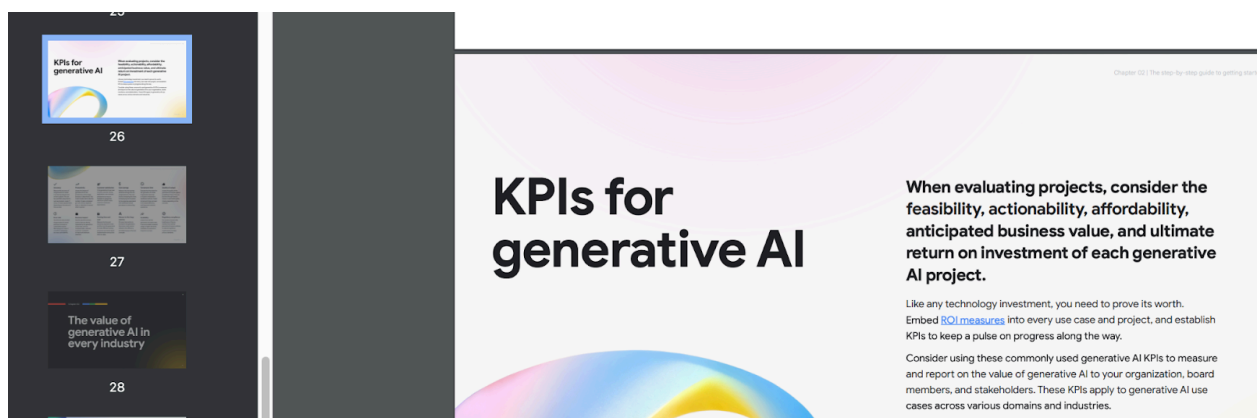


- Feasibility
- Actionability
- Affordability
- Anticipated business value
- Return on investment

[https://storage.cloud.google.com/pdf-docs-367bc/exec\\_guide\\_gen\\_ai.pdf#:~:text=Host%20a%20hackathon,accuracy%20and%20reliability](https://storage.cloud.google.com/pdf-docs-367bc/exec_guide_gen_ai.pdf#:~:text=Host%20a%20hackathon,accuracy%20and%20reliability).

 Agent (OK)  
Playbook

來源 PDF 文件。



問題：

What are the priority use cases in Retail and CPG in the exec guide?



What are the priority use cases in Retail and CPG in the exec guide?



pdf-docs  
Tool

pdf-docs  
Action

1  
Input parameters

1  
Output parameters



The priority use cases in Retail and CPG in the exec guide are:



- Creative assistance
- Conversational commerce
- Customer service automation
- New product development

[https://storage.cloud.google.com/pdf-docs-367bc/exec\\_guide\\_gen\\_ai.pdf#:~:text=Creative%20assistance%20Empower,full%20story%2033](https://storage.cloud.google.com/pdf-docs-367bc/exec_guide_gen_ai.pdf#:~:text=Creative%20assistance%20Empower,full%20story%2033)



Agent (OK)  
Playbook



來源 PDF 文件。

**Retail and CPG**

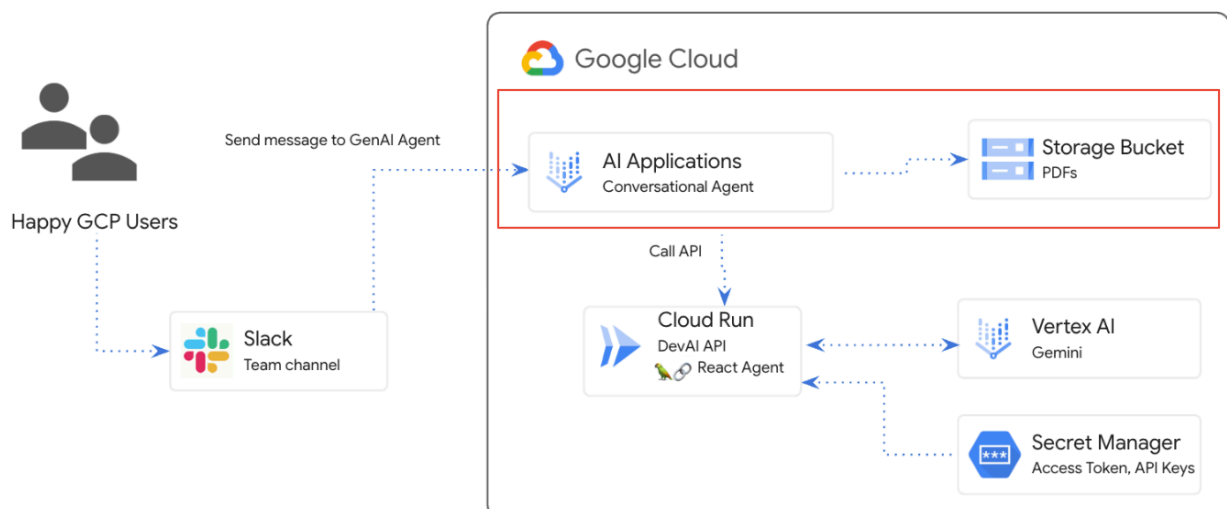
Insight

**82%**  
of retail organizations consider customer service automation to be valuable<sup>5</sup>

**Priority use cases**

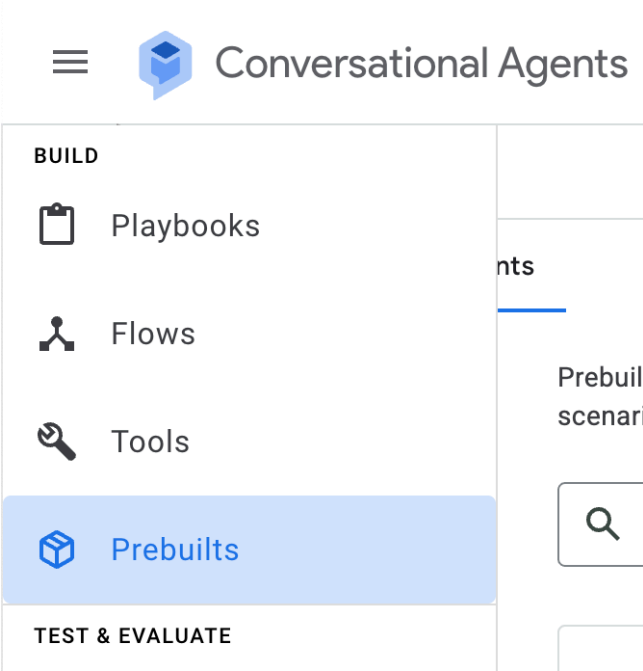
- Creative assistance**  
Empower retail creative teams to create bespoke images and creative content for campaigns and editorial placements, and enable 1:1 personalization.
- Conversational commerce**  
Interactively address queries, provide recommendations, and engage with customers in real time to help them make shopping decisions (for example, "Sure, here are some dresses in your size and style you may like, and here are influencer images for style inspiration").
- Customer service automation**  
Streamline customer service with conversation summaries and task automation.
- New product development**  
Enhance internal consumer research with easy querying, summarization, and insight generation. Create copy concepts and claims for further testing, and visual concepts for product and packaging designs.

恭喜！現在代理可以根據 PDF 文件提供有根據的答案。



## 10. 預先建構的代理程式

接著，您要從左側選單探索預先建構的代理程式。



選取其中一個代理程式並部署。瞭解代理程式的設定、指令和工具。

## Prebuilt

Flow components **Agents**

Prebuilt Generative Agents can be used to accelerate development for new Agents. They showcase potential solutions for common tasks and scenarios, and can help highlight best practices. Prebuilt Agents are currently only available in English. [Learn more](#)

 Search

### Movie Expert <sup>[Featured]</sup>

Beginner Starts with Playbook



Can answer a wide range of movie-related questions.

#### Features

- Generative AI
- BigQuery Integration

#### Required Settings

- Cloud Run/Cloud Functions
- BigQuery

### Airline Support <sup>[Featured]</sup>

Intermediate Starts with Playbook



Simplifies flight booking using geolocation, and displays flight information using rich content.

#### Features

- Generative AI
- Geolocation

#### Required Settings

- Cloud Run/Cloud Functions
- Firestore

### Shopping Assistant <sup>[Featured]</sup>

Advanced Starts with Playbook



Provides retail product queries, ordering, order status, and order cancellation.

#### Features

- Generative AI

#### Required Settings

- Vertex AI Search for Retail & Recommendations API
- Cloud Run/Cloud Functions

### Small talk

Beginner Starts with Flow



Customize and personalize your agent with simple questions and responses.

#### Features

- Pages
- Route groups

### Travel: baggage claim

Beginner Starts with Flow



Create or check the status of a claim related to lost, delayed, or damage baggage.

#### Features

- Regexp entities

### Travel: car rental

Beginner Starts with Flow



Start a new car rental reservation.

#### Features

- System entities
- Custom entities
- Conditional routes

## 11. 恭喜！

恭喜，您已完成本實驗室！

### 涵蓋內容：

- 如何建立及部署 Conversational Agents
- 如何為以 Cloud Run 應用程式為基礎的代理程式新增工具
- 如何將代理程式整合至 Slack 工作區
- 如何設定資料儲存庫，以便對 PDF 文件進行問答

### 後續步驟：

- 查看 Conversational Agents 的[最佳做法](#)

### 清除所用資源

如要避免系統向您的 Google Cloud 帳戶收取本教學課程所用資源的費用，請刪除含有相關資源的專案，或者保留專案但刪除個別資源。

### 刪除專案

如要避免付費，最簡單的方法就是刪除您為了本教學課程所建立的專案。

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